2.6 Quiz: Trigonometry — Markscheme

Version (a)

Total: 34 marks. Accuracy marks need 3 s.f., the exact value, or the calculator display with an ellipsis. In Problems 1 and 3 the calculator must be in radian mode: a value worked in degree mode earns the M and substitution marks if the method is right, and A0 for the answer. Problem order revised 2026-10-28: the former Problem 2 (two boats) was replaced by the unit-circle problem.

1. [Maximum mark: 8]

   (a) $BC^2 = 6^2 + 10^2 - 2(6)(10)\cos 1.1$ **M1A1**

   $BC^2 = 81.5684\ldots$

   $BC = 9.03\,\text{cm} \quad (9.03152\ldots)$ **A1**

   Degree mode gives $BC = 4.00$*: M1A1A0.*

   (b) Area $= \frac{1}{2}(6)(10)\sin 1.1$ **M1**

   Area $= 26.7\,\text{cm}^2 \quad (26.7362\ldots)$ **A1**

   Degree mode gives 0.576*: M1A0.*

   (c) $\dfrac{\sin A\hat{C}B}{6} = \dfrac{\sin 1.1}{9.03152\ldots}$ **M1A1**

   $A\hat{C}B = 0.634 \quad (0.633617\ldots)$ **A1**

   M1 for the sine rule, A1 for pairing $AB = 6$ *with* $A\hat{C}B$*. Wrong method: pairing* 10 *with* $A\hat{C}B$ *gives* 1.41*, which is* $A\hat{B}C$*: M1A0A0. The answer must be in radians:* $36.3°$ *earns M1A1A0. Accept the cosine rule. FT from (a).*

2. [Maximum mark: 9]

(a) $A\hat{C}B = 180° - 68° - 47° = 65°$ **A1**

(b) $\dfrac{AC}{\sin 47°} = \dfrac{40}{\sin 65°}$ **M1A1**

$AC = 32.3\,\text{m}$ $\quad (32.2783\ldots)$ **A1**

M1 for the sine rule, A1 for pairing AC with $47°$ and 40 with $65°$. Wrong method: pairing AC with $68°$ gives 40.9, which is BC: M1A0A0. FT from (a).

(c) $\text{Area} = \frac{1}{2}(40)(32.2783\ldots)\sin 68°$ **M1**

$\text{Area} = 599\,\text{m}^2$ $\quad (598.559\ldots)$ **A1**

M1 for two sides with the angle between them. Wrong method: using $\sin 65°$ with AB and AC gives 585: M0A0. FT from (b).

(d) $\frac{1}{2}(24)(35)\sin\theta = 320$ **M1**

$\sin\theta = \frac{16}{21}$, $\quad$ acute value $49.6°$ **(A1)**

$\theta = 180° - 49.6324\ldots° = 130°$ $\quad (130.367\ldots)$ **A1**

Giving only the acute value $49.6°$ earns M1A1A0. A1 for 130 (3 s.f.) or the display with an ellipsis; 130.4 with no ellipsis earns A0.

3. [Maximum mark: 10]

(a) $a = \cos\frac{\pi}{3} = \frac{1}{2}$ **A1**

$b = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$ **A1**

Accept 0.5 *for* a. b *must be exact:* 0.866 *earns A0. Wrong method: swapping sine and cosine (*$a = \frac{\sqrt{3}}{2}$, $b = \frac{1}{2}$*): A0A0.*

(b) $225° = 225 \times \frac{\pi}{180} = \frac{5\pi}{4}$ **A1**

Accept $\frac{225\pi}{180}$ *or* 1.25π. 3.93 *earns A0 (not a multiple of* π*).*

(c) $c = \cos 225° = -\frac{\sqrt{2}}{2}$ **A1**

$d = \sin 225° = -\frac{\sqrt{2}}{2}$ **A1**

Accept $-\frac{1}{\sqrt{2}}$. *A1 for each value with its sign;* $\frac{\sqrt{2}}{2}$ *with no minus sign earns A0 for that value. Decimals (*-0.707*) earn A0.*

(d) $k^2 + 0.6^2 = 1$ **M1**

$k = 0.8$ **A1**

M1 for $x^2 + y^2 = 1$ *(Pythagoras in the unit circle), or for* $k = \cos(\arcsin 0.6)$. *Accept* $\frac{4}{5}$. *The negative root* -0.8 *belongs to S: giving* $k = \pm 0.8$ *as the final answer earns M1A0. Wrong method:* $k = 1 - 0.6 = 0.4$*: M0A0.*

(e) $\alpha = \arcsin 0.6 = 0.644 \quad (0.643501\ldots)$ **A1**

$\beta = \pi - \alpha$ **M1**

$\beta = 2.50 \quad (2.49809\ldots)$ **A1**

M1 for $\pi - \alpha$*, or for the symmetry of S and R about the y-axis. Answers in degrees (*$36.9°$, $143°$*) earn A0, M1, A0. Wrong methods:* $\pi + \alpha = 3.79$ *or* $2\pi - \alpha = 5.64$ *(these have* $\sin\theta = -0.6$*): M0A0. Accept* 2.50 *from the rounded* $\alpha = 0.644$. *FT from the candidate's* α.

4. [Maximum mark: 7]

(a) $\tan 32° = \dfrac{FT}{35}$ **M1**

$FT = 35 \tan 32° = 21.9\,\text{m}$ $\quad (21.8704\ldots)$ **A1**

Wrong methods: $35 \sin 32° = 18.5$ *or* $\frac{35}{\tan 32°} = 56.0$ *earn M0A0.*

(b) $AT = \dfrac{35}{\cos 32°}$ **M1**

$AT = 41.3\,\text{m}$ $\quad (41.2712\ldots)$ **A1**

Accept Pythagoras with 35 *and* FT*, or* $\frac{FT}{\sin 32°}$*. Wrong method:* $\frac{35}{\sin 32°} = 66.0$ *earns M0A0. FT from (a).*

(c) $BF = 35 - 15 = 20$ **A1**

$\tan\theta = \dfrac{21.8704\ldots}{20}$ **M1**

$\theta = 47.6°$ $\quad (47.5578\ldots)$ **A1**

M1 for $\tan\theta = \frac{height}{their\ BF}$*. Wrong method: using* 15 *as the horizontal distance gives* 55.6°*: A0M1A0. FT from (a).*

Version (b)

Total: 34 marks. Accuracy marks need 3 s.f., the exact value, or the calculator display with an ellipsis. In Problem 2 the calculator must be in radian mode: a value worked in degree mode earns the M and substitution marks if the method is right, and A0 for the answer.

1. [Maximum mark: 8]

(a) $A\hat{B}T = 180° - 40° = 140°$, so $A\hat{T}B = 180° - 23° - 140° = 17°$ **A1**

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Accept the exterior angle: $40° = 23° + A\hat{T}B$. *A1 needs* $140°$ *or the exterior-angle reason seen; restating* $17°$ *earns A0.*

(b) $\dfrac{BT}{\sin 23°} = \dfrac{50}{\sin 17°}$ **M1A1**

$BT = 66.8\,\text{m}$ $\quad (66.8209\ldots)$ **A1**

M1 for the sine rule in triangle ABT, A1 for pairing BT with $23°$ *and* 50 *with* $17°$. *Wrong method: pairing BT with* $140°$ *(or* $40°$*) gives* 109.9*, which is AT: M1A0A0.*

(c) $FT = 66.8209\ldots \times \sin 40°$ **M1**

$FT = 43.0\,\text{m}$ $\quad (42.9516\ldots)$ **A1**

Accept $AT \sin 23°$ *with* $AT = 109.926\ldots$. *Accept* 42.9 *from the rounded* $BT = 66.8$. *Wrong method:* $BT \cos 40° = 51.2$ *(the distance BF): M0A0. FT from (b).*

(d) $BF = 66.8209\ldots \times \cos 40°$ **M1**

$BF = 51.2\,\text{m}$ $\quad (51.1878\ldots)$ **A1**

Accept $\frac{FT}{\tan 40°}$*, Pythagoras with BT and FT, or* $AT \cos 23° - 50$. *Wrong method:* $BT \sin 40° = 43.0$ *(the height): M0A0. FT from (b) or (c).*

2. [Maximum mark: 8]

(a) $\dfrac{\sin P\hat{R}Q}{12} = \dfrac{\sin 0.7}{9}$ **M1**

$\sin P\hat{R}Q = 0.858956\ldots$

$P\hat{R}Q = 1.03 \quad (1.03322\ldots)$ **A1**

$P\hat{R}Q = \pi - 1.03322\ldots = 2.11 \quad (2.10836\ldots)$ **A1**

Both values are possible because $0.7 + 2.10836\ldots < \pi$. *Wrong methods: giving only* 1.03 *earns M1A1A0; degree mode (*$\sin P\hat{R}Q = 0.0163$*) earns M1A0A0; answers in degrees (*$59.2°$, $121°$*) earn M1A0A0.*

(b) $P\hat{Q}R = \pi - 0.7 - 2.10836\ldots = 0.333 \quad (0.333229\ldots)$ **A1**

Accept 0.332 *from the rounded* $P\hat{R}Q = 2.11$. *Wrong method: using the acute value gives* 1.41*: A0. FT from (a).*

(c) $\dfrac{PR}{\sin 0.333229\ldots} = \dfrac{9}{\sin 0.7}$ **M1**

$PR = 4.57\,\text{cm} \quad (4.56967\ldots)$ **A1**

Accept the cosine rule $9^2 = 12^2 + PR^2 - 2(12)(PR)\cos 0.7$*: it gives* $PR = 4.57$ *or* 13.8*, and the obtuse case is the smaller root; giving* 13.8 *(the acute case) earns M1A0. Accept* 4.55 *from the rounded* $P\hat{R}Q = 2.11$. *FT from (b).*

(d) Area $= \frac{1}{2}(12)(9)\sin 0.333229\ldots$ **M1**

Area $= 17.7\,\text{cm}^2 \quad (17.6631\ldots)$ **A1**

Accept $\frac{1}{2}(12)(PR)\sin 0.7$. *Accept* 17.6 *from the rounded* $P\hat{R}Q = 2.11$. *Wrong method:* $\frac{1}{2}(12)(9)\sin 0.7 = 34.8$ *(0.7 is not between these sides): M0A0. FT from (b) or (c).*

Feedback: in this problem rounding an angle early moves the later answers. Store the full values in the calculator.

3. [Maximum mark: 9]

(a) $AC^2 = 8^2 + 5^2 - 2(8)(5)\cos 125°$ **M1A1**

$AC^2 = 134.886\ldots$

$AC = 11.6\,\text{cm}$ $\quad (11.6140\ldots)$ **A1**

M1 for the cosine rule, A1 for the correct substitution. Wrong method: a sign error $(89 - 45.9)$ *gives* 6.57*: M1A0A0.*

(b) $\cos A\hat{D}C = \dfrac{7^2 + 12^2 - 11.6140\ldots^2}{2(7)(12)}$ **M1A1**

$\cos A\hat{D}C = 0.345915\ldots$

$A\hat{D}C = 69.8°$ $\quad (69.7622\ldots)$ **A1**

M1 for the cosine rule for an angle, A1 for AC *as the side opposite* $A\hat{D}C$*. Accept* $69.6°$ *from the rounded* $AC = 11.6$*. Wrong method: using* $AD = 12$ *as the opposite side gives* $75.8°$*, which is* $A\hat{C}D$*: M1A0A0. FT from (a).*

(c) Area $ABC = \frac{1}{2}(8)(5)\sin 125° = 16.3830\ldots$

Area $ACD = \frac{1}{2}(7)(12)\sin 69.7622\ldots° = 39.4071\ldots$ **M1**

Area $ABCD = 16.3830\ldots + 39.4071\ldots$ **M1**

Area $ABCD = 55.8\,\text{cm}^2$ $\quad (55.7901\ldots)$ **A1**

First M1 for a correct area expression for either triangle (two sides and the angle between them). Second M1 for adding the two triangles. Wrong method: using $125°$ *in triangle* ACD *as well gives* 50.8*: M1M1A0. FT from (b).*

4. [Maximum mark: 9]

(a) $6 + p + 12 + q + 3 = 40$, so $p + q = 19$ **A1**

$$\frac{15(6) + 25p + 35(12) + 45q + 55(3)}{40} = 31.25$$ **M1**

$675 + 25p + 45q = 1250$, so $5p + 9q = 115$

$p = 14$ **A1**

$q = 5$ **A1**

M1 for the mean equation using the class midpoints $15, 25, \ldots, 55$*. Accept solving the system on the GDC. Wrong method: lower class boundaries (*$10, 20, \ldots$*) give* $p = 4$*,* $q = 15$*: A1M0A0A0.*

(b) $\sigma = 11.1$ minutes $\quad (11.1102\ldots)$ **M1A1**

M1 for the midpoints with their frequencies entered in the GDC (may be implied by a correct answer). Wrong method: the sample value $s_x = 11.3$ *earns M1A0. FT from (a).*

(c) new time $= 60(t + 2) = 60t + 120$ seconds **(M1)**

new mean $= 60(31.25 + 2) = 1995\,\text{s}$ **A1**

new standard deviation $= 60 \times 11.1102\ldots = 667\,\text{s} \quad (666.614\ldots)$ **A1**

The penalty moves the mean but not the spread; the conversion to seconds multiplies both by 60*. Accept* 666 *from the rounded* $\sigma = 11.1$*. Wrong methods: adding* 120 *to the standard deviation (*787*), or adding the* 2 *minutes after converting (mean* 1877*): A0 for that value. FT from (b).*